SCIENCE 5 — QUARTER 2, MODULE 3: ANG IKOT NG TUBIG (THE WATER CYCLE)

Self-Learning Module · For classroom use

WHAT I NEED TO KNOW

This module was designed to help you understand how water moves through our environment. After going through this module, you are expected to: (1) describe the processes of the water cycle; (2) explain the role of the sun as the main source of energy of the cycle; and (3) give real-life examples of evaporation, condensation, and precipitation.

WHAT IS THE WATER CYCLE?

Water on Earth is always on the move. The water cycle, also called the hydrologic cycle, describes how water travels from the oceans to the sky, falls back onto the land, and returns once more to the sea. There is no beginning and no end — the same water has been recycled for billions of years.

STAGE 1: EVAPORATION

When the sun heats the surface of oceans, lakes, and rivers, liquid water changes into water vapor and rises into the air. Plants join in as well: they release water vapor through tiny openings in their leaves, in a process called transpiration.

STAGE 2: CONDENSATION

As water vapor rises, the surrounding air becomes colder. The vapor cools and turns back into tiny liquid droplets that gather around specks of dust. Millions of these droplets together form the clouds that we see in the sky. Fog is simply a cloud that forms near the ground.

STAGE 3: PRECIPITATION

When the droplets inside a cloud grow too large and heavy to stay afloat, they fall back to Earth. Depending on the temperature, precipitation may arrive as rain, snow, sleet, or hail. In the Philippines, we experience it most often as rain, especially during the habagat season.

STAGE 4: COLLECTION

The fallen water gathers in rivers, lakes, and oceans, or soaks into the soil to become groundwater. From these collecting places, the sun heats the water once more — and the cycle begins all over again.

WHAT'S MORE — ACTIVITY 1

On a separate sheet of paper, draw the water cycle and label each stage. Then list two examples of evaporation or condensation that you can observe in your own home, such as drying laundry or droplets forming on a cold glass of water.